

Task 1: Research & Documentation (Deadline: Jan 21 11:59pm)

Research and create a one-page summary on the following:

- 1. What is cybersecurity, and why is it important?**
- 2. Five real-world cyberattacks and how they happened.**
- 3. Difference between HTTP and HTTPS.**
- 4. Explanation of AES and RSA encryption with simple examples.**

1. What is Cybersecurity, and Why is it Important?

Cybersecurity refers to the practice of protecting systems, networks, and data from digital attacks, unauthorized access, damage, or theft. It ensures the confidentiality, integrity, and availability of information.

In today's interconnected world, cybersecurity is crucial due to the increasing reliance on digital infrastructures and the rising sophistication of cyber threats.

Effective cybersecurity measures safeguard personal data, protect organizational assets, and maintain national security.

2. Five Real-World Cyberattacks and How They Happened

- **Stuxnet Worm (2010):** A sophisticated malware targeting Iran's nuclear facilities, causing physical damage to centrifuges by exploiting vulnerabilities in industrial control systems.
- **Equifax Data Breach (2017):** Exploited a vulnerability in a web application framework, leading to the exposure of personal information of approximately 147 million people.
- **NotPetya Attack (2017):** Initially spread through a Ukrainian accounting software, this malware caused widespread disruption by encrypting data and demanding ransom, affecting global businesses.
- **Marriott International Data Breach (2018):** Unauthorized access to the Starwood guest reservation database led to the compromise of personal information of approximately 500 million guests.
- **Twitter Bitcoin Scam (2020):** High-profile Twitter accounts were hijacked through a coordinated social engineering attack, promoting a cryptocurrency scam.

Recent Cyber Attacks in India:

- **AIIMS Ransomware Attack (2022):** A major ransomware attack disrupted the All India Institute of Medical Sciences (AIIMS) in Delhi, affecting the hospital's digital services for weeks. Hackers demanded cryptocurrency ransom, compromising patient data and hospital operations.

- **Domino's India Data Breach (2021):** Sensitive customer information, including names, phone numbers, and addresses, was leaked on the dark web. The breach exposed data of millions of users.
- **Mobikwik Data Breach (2021):** Personal data of over 100 million users, including KYC details and financial records, was leaked online after attackers gained access to the payment platform's database.
- **Air India Data Breach (2021):** Hackers accessed and leaked the data of around 4.5 million passengers, including credit card information and travel details, through a cyberattack on the airline's third-party data processor.
- **SpiceJet Ransomware Attack (2022):** The airline's IT systems were targeted, causing significant delays and disruptions in operations. The attack highlighted vulnerabilities in critical infrastructure systems.

3. Difference Between HTTP and HTTPS

- **HTTP (HyperText Transfer Protocol):**
 - HTTP is the foundational protocol used to transfer data on the web. However, data transmitted via HTTP is in plain text, making it vulnerable to interception by hackers.
 - Commonly used in non-sensitive interactions, such as accessing general web pages without requiring login or personal information.
- **HTTPS (HTTP Secure):**
 - HTTPS is an extension of HTTP that incorporates encryption via SSL (Secure Sockets Layer) or TLS (Transport Layer Security) protocols. This ensures that data transferred between a user's browser and the server is encrypted, protecting it from eavesdropping or tampering.
 - Essential for transactions requiring sensitive information, such as online banking, shopping, or filling out personal forms.
 - Provides authentication by verifying the identity of the website using digital certificates. This ensures users connect to legitimate sites, mitigating the risk of phishing attacks.

Key Differences:

- **Encryption:** HTTPS encrypts data, while HTTP does not.
- **Security Indicators:** HTTPS-enabled websites display a padlock icon and use "https://" in the URL.
- **Performance:** HTTPS may introduce a slight delay due to encryption processes, but modern systems are optimized for speed.

4. Explanation of AES and RSA Encryption with Simple Examples

- **AES (Advanced Encryption Standard):**

- A symmetric encryption algorithm that uses the same key for both encryption and decryption.
- Example: If Seja and Anand share a secret key, Seja can encrypt a message using AES, and Anand can decrypt it using the same key. For instance, if the key is “1234” and the message is “Hello,” the encrypted output might look like “A1B2C3,” which Anand can decrypt back to “Hello” using the key.
- **RSA (Rivest-Shamir-Adleman):**
 - An asymmetric encryption algorithm that uses a pair of keys—a public key for encryption and a private key for decryption.
 - Example: If Anand wants to send a secure message to Seja, he can encrypt it using Seja’s public key. Only Seja can decrypt it using her private key. For instance, Anand encrypts “Hi Seja” with Seja’s public key, resulting in “X9Y8Z7.” When Seja receives it, she uses her private key to decrypt it back to “Hi Seja.”